



Overview

MOVIELENS 1M · AGE GROUP ANALYSIS



MOVIELENS 1M — AGE GROUP ANALYTICS



Total Ratings

1000209

Users

6040

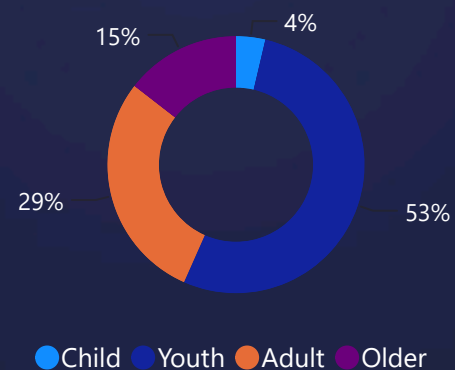
Movies

3706

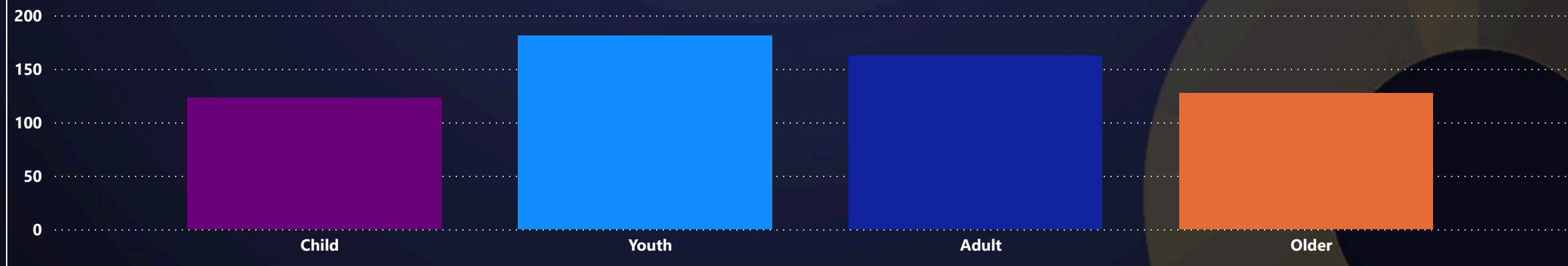
Avg Rating

4

User base share by age group



Average ratings per user by age group





Genre

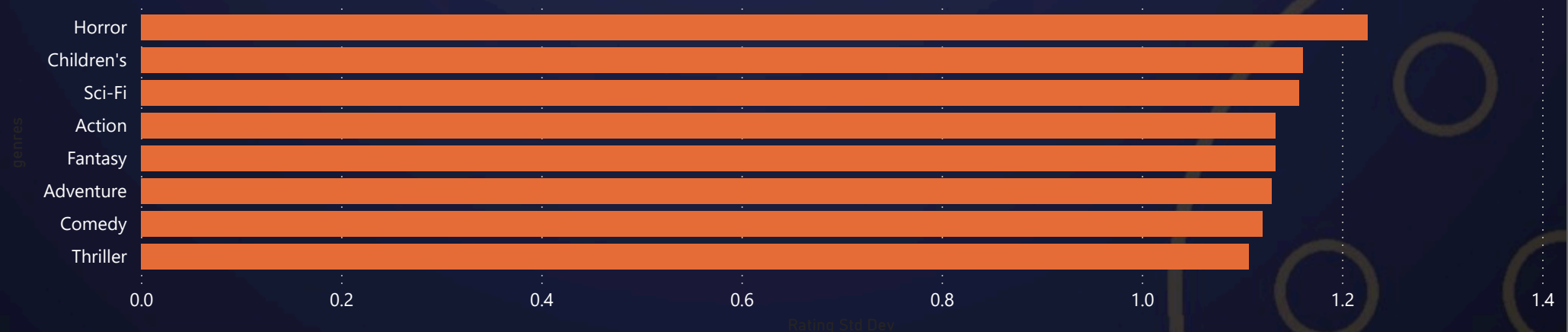
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Avg rating by genre and age grou

genres	Child	Youth	Adult	Older	Total
Horror	3.25	3.19	3.27	3.19	3.22
Children's	3.24	3.37	3.52	3.58	3.42
Fantasy	3.32	3.42	3.50	3.57	3.45
Sci-Fi	3.48	3.44	3.50	3.54	3.47
Adventure	3.45	3.43	3.52	3.64	3.48
Action	3.51	3.45	3.54	3.61	3.49
Comedy	3.50	3.48	3.57	3.65	3.52
Thriller	3.55	3.52	3.62	3.71	3.57
Romance	3.62	3.54	3.66	3.78	3.61
Western	3.58	3.57	3.68	3.76	3.64
Musical	3.57	3.60	3.73	3.83	3.67

Most polarizing genres (rating std dev)





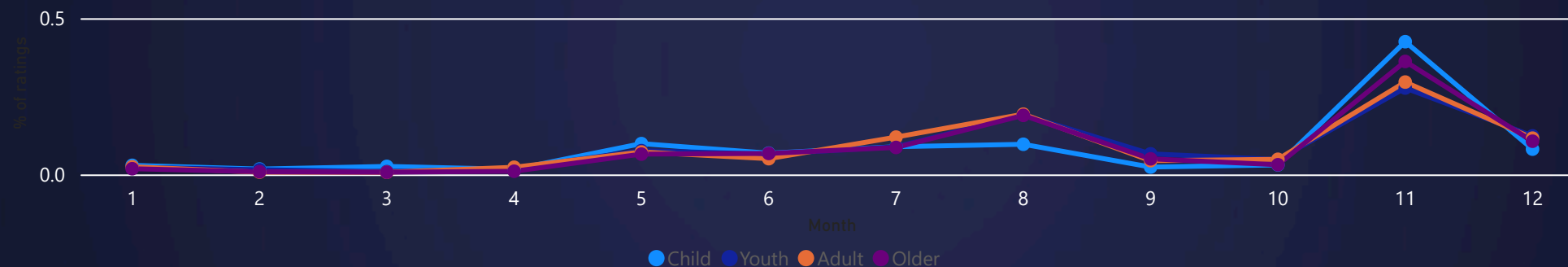
Timing

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IsFirstDay

All

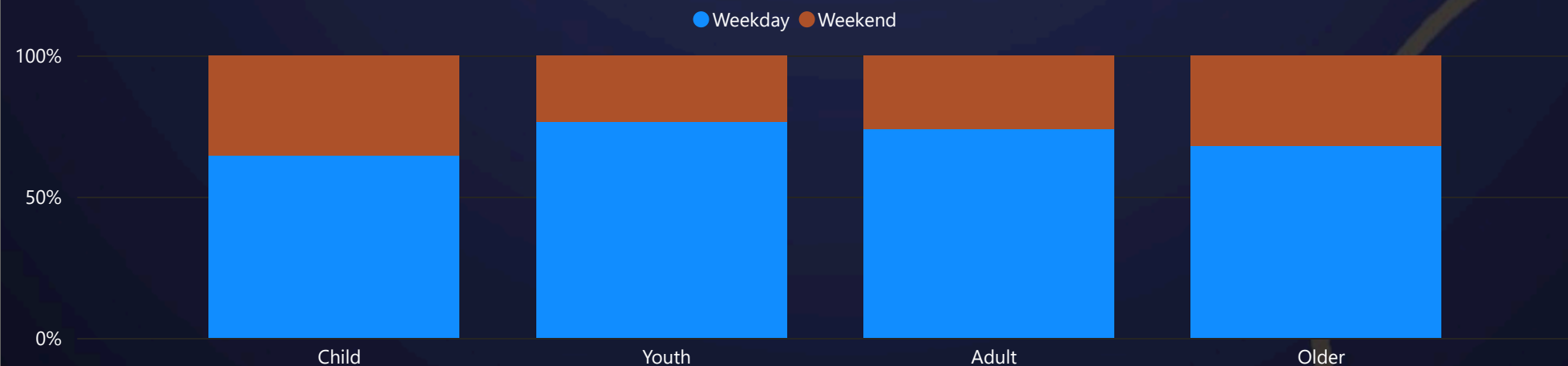
Monthly rating activity per group



66.0%

Share of ratings made on a user's first day of activity. High values mean the monthly pattern above is partly a signup-wave artifact, not pure calendar seasonality

Weekday vs weekend split by age group





Stats

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Child gender split, Children's genre



3.53



2.99

Older vs Child, generosity

Older vs Child generosity



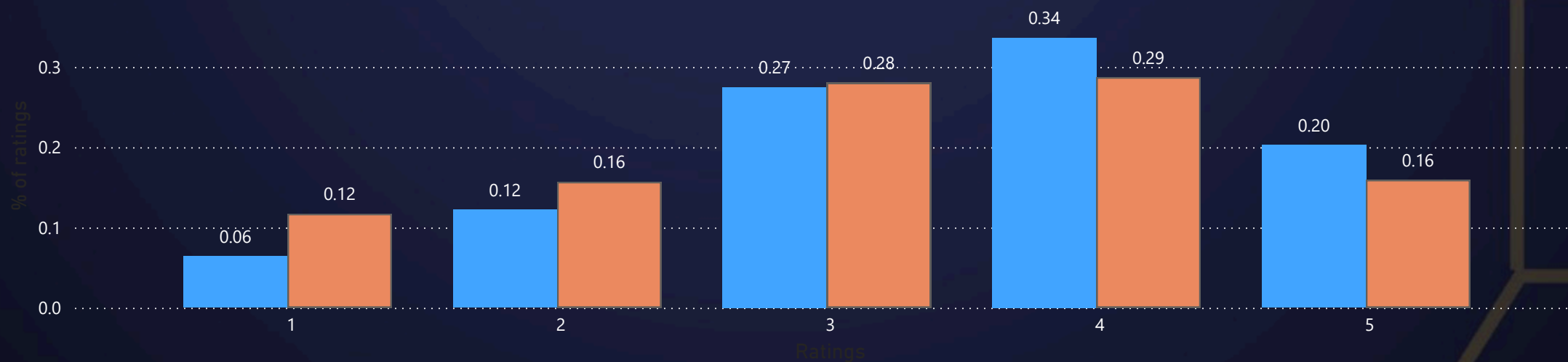
3.82



3.62

Rating distribution: Horror vs Action

● Action ● Horror



i Note: Horror's std dev is highest across every age group, but the distribution is right-skewed with a heavier low-rating tail rather than a true bimodal split.

Significance testing on the two most interesting gaps



Recommender

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CF Model RMSE

0.99

Baseline RMSE

1.11

User Label

Adult group, user 30



Note: Predictions from an item-based CF model. Beats baseline by ~10% RMSE

Top Rated by User

Title	Rating
Bamboozled (2000)	5.00
Double Life of Veronique, The (La Double Vie de Veronique) (1991)	5.00
E.T. the Extra-Terrestrial (1982)	5.00
Schindler's List (1993)	5.00
Usual Suspects, The (1995)	5.00

Predicted rating

Title	Rating
Kid in King Arthur's Court, A (1995)	3.67
Wing Commander (1999)	3.66
Solo (1996)	3.65
Star Wars: Episode IV - A New Hope (1977)	3.65
Star Wars: Episode VI - Return of the Jedi (1983)	3.65